NG JING YAO

Software Engineering Graduate | Full-Stack Development | AI Application Development

016-9568231 | jingyaong251104@gmail.com

LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Software Engineering graduate with hands-on experience developing full-stack and AI-powered applications using Python, Flask, React.js, Docker, Retrieval-Augmented Generation (RAG), LLM APIs, and LLM tool calling.

EXPERIENCE

Intern

YTL Construction Sdn Bhd | Jan 2026 - Jul 2026

- Developed a full-stack RAG chatbot covering 20+ internal IT and HR documents using Flask, LangChain, ILMU through OpenAI-compatible LLM APIs, and hybrid retrieval with ChromaDB, BM25, and cross-encoder reranking.
- Built the frontend using SharePoint Framework (SPFx), React.js, and TypeScript, supporting streaming responses, conversation memory, feedback collection, and error reporting.
- Created a document ingestion pipeline for Markdown/PDF sources, including hierarchical chunking, metadata tracking, abbreviation extraction, embeddings, and incremental re-indexing.
- Enhanced query processing with conversation-aware rewriting, language translation, abbreviation expansion, and multi-query generation to improve retrieval relevance.
- Implemented 5 LLM tool-calling workflows with backend validation, structured results, human-in-the-loop confirmation, and guardrails for source-grounded answers and sensitive information.
- Deployed separate containerized IT and HR chatbot services on Linux using Docker and collaborated with approximately 20 internal users to improve knowledge coverage and answer reliability.

Software Development Intern

YYC GST Consultants Sdn Bhd | Nov 2023 - Feb 2024

- Assisted in developing web application modules using Vue.js and Laravel, contributing to frontend and backend functionality.
- Worked with Laravel backend APIs to retrieve data from multiple database tables for use in the application.
- Developed dynamic tables using Vue.js to display backend query results in a clear, user-friendly format.
- Collaborated with the development team using GitHub for version control and frontend-backend integration.
- Participated in testing and debugging to identify and resolve application issues before deployment.

PROJECTS

EasyHealthy – Diet and Fitness Tracker Mobile App

Final Year Project (FYP)

- Developed meal-tracking features using Flutter and Cloud Firestore, integrating USDA FoodData Central search, portion adjustments, meal logging, and custom food management.
- Implemented a hybrid recipe recommendation system combining nutritional matching with user preferences derived from saved recipes and meal logs.
- Built daily meal plan generation for breakfast, lunch, dinner, and snacks using calorie and macronutrient targets, repeat-recipe penalties, and side-dish selection.
- Contributed to nutrition progress visualizations, calorie-goal updates, and shared home and profile features as part of a two-person final year project.

TECHNICAL SKILLS

Backend: Python, Flask, Java, PHP, Laravel, REST APIs
Frontend: React.js, TypeScript, JavaScript, Vue.js, Flutter, Dart, SharePoint Framework (SPFx), HTML, CSS
Databases & Data: SQL, MySQL, Firebase, Cloud Firestore
Cloud & Deployment: Amazon Web Services (AWS), Docker, Linux, Internet Information Services (IIS)
AI/LLM: Retrieval-Augmented Generation (RAG), prompt engineering, LLM APIs, LLM tool calling, LangChain, ChromaDB, BM25, cross-encoder reranking
Tools: Git, GitHub, XAMPP, Codex, Claude Code

EDUCATION

Bachelor of Software Engineering

Tunku Abdul Rahman University of Management and Technology (TARUMT) | Jul 2024 - Aug 2026

CGPA: 3.73 | Dean's List: Nov 2024 | President's List: Jan 2026

Diploma in Information Technology

Tunku Abdul Rahman University of Management and Technology (TARUMT) | Jun 2022 - Jun 2024

CGPA: 3.79

LANGUAGES

English (Proficient) | Chinese (Native) | Malay (Intermediate)